FPT UNIVERSITY HO CHI MINH CITY CAMPUS

Housing Price Prediction in Ho Chi Minh City Using Machine Learning Techniques

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Abstract

This paper applies and adapts the machine learning pipeline proposed by Truong et al. (2020) to the residential real-estate market of Ho Chi Minh City (HCMC), Vietnam. Working from a dataset of 24,323 property listings, rigorous preprocessing — missing-value imputation, IQR-based outlier removal, dimension-string parsing, and ward extraction — yields a clean dataset of 21,595 records with 9 engineered features. Five models are trained and evaluated using Root Mean Squared Logarithmic Error (RMSLE): Random Forest (RF), XGBoost, LightGBM, Hybrid Regression (equal-weight average), and Stacked Generalization (RF + LightGBM base learners, XGBoost meta-learner, 5-fold CV). Without neighbourhood-level context, the best test RMSLE is 0.3479 (Hybrid Regression). After introducing a communityAverage proxy feature — the median price per m² of each ward — the best test RMSLE improves to 0.3312 (Hybrid Regression), confirming the critical predictive value of local neighbourhood pricing. GridSearchCV tuning of XGBoost (learning_rate=0.1, max_depth=6, n_estimators=400) further reduces its standalone test RMSLE from 0.3553 to 0.3492. Results are consistent with the reference paper: ensemble methods outperform individual models, and Stacked Generalization achieves the lowest overfitting gap.

Keywords: *Housing Price Prediction; Machine Learning; Random Forest; XGBoost; LightGBM; Stacked Generalization; Ho Chi Minh City*

1. Introduction

The real estate market of Ho Chi Minh City (HCMC) is one of the fastest-growing in Southeast Asia, yet it exhibits extreme spatial heterogeneity. For instance, median prices per square metre in central wards such as Phường Sài Gòn (648.5 triệu VND/m²) can exceed those in peripheral wards by a factor of ten or more. Currently, prospective homebuyers primarily rely on manual comparable-sales analysis or depend entirely on information provided by real estate brokers. This reliance introduces a significant risk: it is exceedingly difficult for buyers to estimate the deviation between the broker's quoted price and the true market value of the property. Furthermore, manual valuation methods are labour-intensive and fail to capture the complex, nonlinear interactions among various property attributes.

To address this challenge, our project implements an automated housing price prediction system to provide an objective valuation metric for both buyers and sellers. Specifically, this paper replicates and adapts the machine learning pipeline proposed by Truong et al. (2020), tailoring it exclusively to the HCMC residential market.

Key adaptations include: (i) a dataset of 24,323 listings instead of 300,000+; (ii) only 8 raw features, necessitating additional feature engineering; (iii) Vietnamese address parsing to extract ward-level location signals; and (iv) construction of a communityAverage proxy to approximate the neighbourhood price context identified as highly predictive in the reference paper. The paper makes three concrete contributions: a cleaned HCMC housing dataset; a comprehensive five-model benchmark; and an empirical quantification of the communityAverage feature's impact.

2. Dataset and Preprocessing

2.1. Raw Data

The raw dataset consists of 24,323 residential property listings scraped from Vietnamese real-estate portals, representing transactions in HCMC in 2025–2026. Each record contains 8 raw columns: Price(VND), Acreage (m^2), Area (free-text dimension string, e.g., "6x30m"), Bedrooms, Floors (5.7% missing), Parking (binary), Address (Vietnamese free text), and District. Table 1 summarises the raw attributes.

Table 1. Raw Dataset Attributes

Attribute	Data Type	Description
Price(VND)	float64	Sale price in Vietnamese Dong
Acreage	int64	Total floor area (m^2)
Area	str	Dimension string, e.g., "6x30m" (8.0% missing)
Bedrooms	int64	Number of bedrooms
Floors	float64	Number of floors (5.7% missing)
Parking	int64	Parking availability (binary: 0/1)
Address	str	Vietnamese address string
District	str	Administrative district (65 unique values)

2.2. Exploratory Data Analysis

The target variable Price(VND) exhibits extreme right skewness (skewness = 36.49), with values ranging from 0 to 6,728.54 tỷ VND. The mean (26.67 tỷ) is more than twice the median (10.80 tỷ), indicating severe right-tail distortion caused by 2,577 IQR outliers (10.6% of the raw data). If fed directly into the models, these outliers would introduce significant noise and cause the algorithms to learn incorrectly. To address this and normalize the distribution, a log1p transformation is applied to the target variable, successfully reducing the skewness to −0.242. The transformation formula is defined as:

$$Y\_new = \log(Price + 1)$$

Fig. 1 displays the raw price histogram (capped at 100 tỷ), the log-transformed distribution, and a boxplot illustrating the extent of outliers. The price distribution for listings below 100 tỷ follows an approximately exponential decay, with the highest concentration in the 5–15 tỷ range (which jointly contains over 40% of listings).

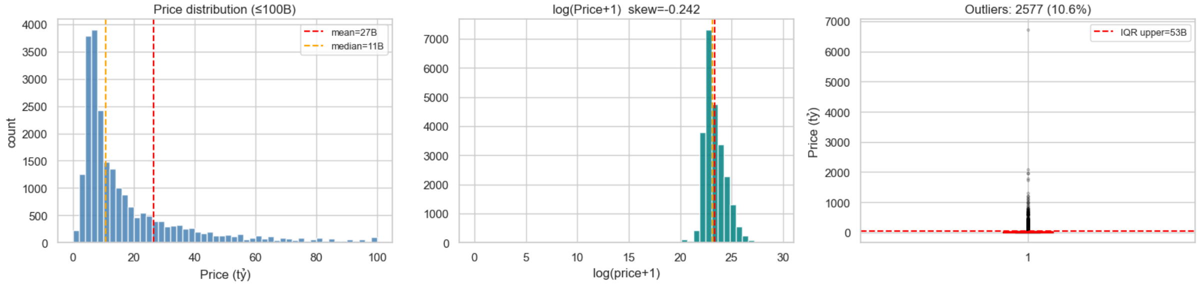

Fig. 1. Price distribution (raw ≤100 tỷ), log(Price+1) distribution (skew = −0.242), and outlier boxplot (IQR upper = 53 tỷ)

Geographic analysis reveals pronounced spatial price gradients. The five highest-priced wards by median price per m² are Phường Sài Gòn (648.5M/m²), Phường Bến Thành (531.8M/m²), Phường Cầu Ông Lãnh (371.4M/m²), Phường Tân Định (351.4M/m²), and Phường Xuân Hòa (350.0M/m²), all located in or adjacent to District 1. Price correlates positively with Acreage (r = 0.642 among non-extreme listings), indicating that property size is the strongest single physical predictor. The Parking binary feature shows a counterintuitive pattern: properties without parking have a higher median price, suggesting central properties command a premium regardless of parking availability.

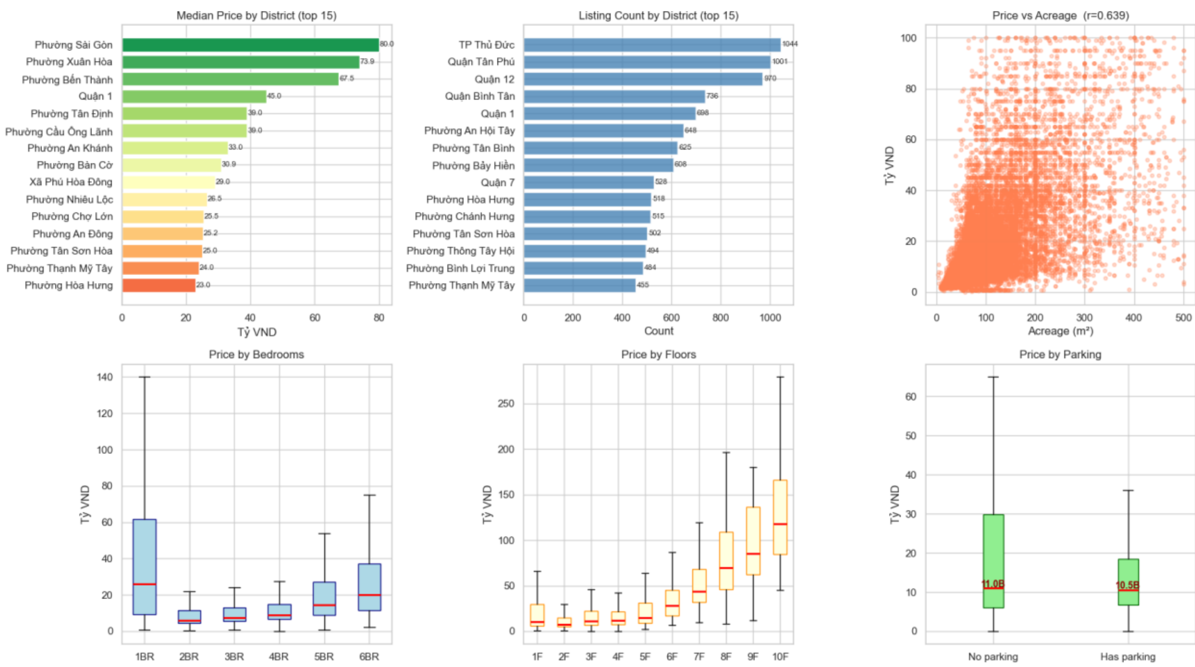

Fig. 2. EDA overview: median price by district (top 15), listing count by district (top 15), price vs. acreage scatter (r = 0.642), and price distributions by bedrooms, floors, and parking

2.3. Preprocessing Pipeline

The preprocessing pipeline follows Truong et al. (2020) in spirit, adapted for HCMC-specific data characteristics. Steps applied sequentially:

- Remove records with Price < 500 triệu VND (likely data entry errors) and records with missing Area (8.0% of raw data).
- Impute missing Floors values (5.7%) with the column median.
- Apply 1st–99th percentile clipping on Price, retaining the range 2.0 tỷ – 201.6 tỷ VND and removing 2.8% of records.

- Parse the Area string (e.g., "6x30m") into two numerical features: width and length. Records with unparseable formats or out-of-range dimensions (width > 50 m or length > 100 m) are dropped.
- Extract Ward (Phường / Xã / Thị trấn) from the Vietnamese Address string using regular expression matching. Records without a parseable ward are dropped.
- Compute frontage_ratio = width / length as a derived shape feature.

After rigorous preprocessing, the final clean dataset is reduced from over 24,000 raw rows to 21,595 records, containing 9 high-quality modelling features. Table 2 summarises the data reduction at each stage.

Table 2. Data Reduction at Each Preprocessing Stage

Step	Records Removed	Records Remaining
Raw data	—	24,323
Remove Price < 500M + missing Area	~1,963	~22,360
1%–99% price percentile clip	~625	~21,735
Area parsing + range filter + Ward extraction	~140	21,595

2.4. Feature Engineering and Target Encoding

The 9 baseline modelling features are: Acreage, Bedrooms, Floors, Parking, width, length, frontage_ratio, District_enc, and Ward_enc. Instead of using traditional One-Hot Encoding, which would cause a massive dimensionality explosion, the two high-cardinality categorical variables — District (65 unique values) and Ward — are encoded via target encoding. Each category is replaced by the median log-price of the training-set records in that specific area. Critically, the encoding mapping is fit exclusively on the training fold (80% of data) before being applied to the test fold (20%), preventing data leakage. Unseen categories are imputed with the global training median.

Furthermore, a second feature set (10 features in total) is constructed by adding a *communityAverage* proxy. This feature represents the median price per m² of all training-set records in the same Ward, acting as a crucial "regional valuation coefficient" to capture neighbourhood-level pricing context, analogous to the highly predictive feature identified by Truong et al. (2020).

3. Model Selection and Implementation

The evaluation metric is Root Mean Squared Logarithmic Error (RMSLE). Since the models predict log(Price + 1) directly, RMSLE is equivalent to RMSE on log-prices, measuring relative (percentage-scale) prediction error:

$$\mathbf{RMSLE} = \sqrt{\left[(1/N)\, \Sigma_i\, (\log(\hat{y}_i + 1) - \log(y_i + 1))^2 \right]}$$

In addition to RMSLE, metrics such as R2 (which indicates the proportion of variance in price explained by the model), Mean Absolute Error (MAE), and Mean Squared Error (MSE) are also recorded to provide a comprehensive evaluation of the actual prediction deviations. All models are implemented in Python using scikit-learn (v1.4), XGBoost (v2.0), and LightGBM (v4.3).

3.1. Random Forest

Random Forest is utilised as a strong baseline model, combining predictions from 900 independently trained decision trees. Hyperparameters follow Truong et al. (2020): *n_estimators=900, max_depth=20, min_samples_split=10*. However, empirical evaluation reveals that this model suffers from severe overfitting on this specific dataset. Consequently, our study places a deeper focus on Boosting and Ensemble models to mitigate this issue.

3.2. Extreme Gradient Boosting (XGBoost)

XGBoost grows trees sequentially to minimise a regularised objective. To optimise performance, we employed GridSearchCV to fine-tune the hyperparameters. The model was trained over 400 epochs *(n_estimators=400)* with a learning_rate of 0.1 and a *max_depth* of 6. *Why adopt this configuration?* While the original reference paper relied on lower learning rates, our experiments on the HCMC dataset demonstrated that a rate of 0.1 yields superior results. This suggests that our data requires a faster, less conservative weight-updating schedule compared to the Beijing dataset.

3.3. Light Gradient Boosting Machine (LightGBM)

LightGBM uses leaf-wise tree growth and Gradient-based One-Side Sampling (GOSS), achieving faster training while maintaining competitive accuracy. The model was configured with a *learning_rate* of 0.15 and 64 iterations *(n_estimators=64)*. Notably, this configuration allows the model to train in just 0.1 seconds — approximately 45 times faster than the Random Forest baseline.

3.4. Hybrid Regression

Hybrid Regression averages the predictions of RF, XGBoost (tuned), and LightGBM with equal weights (33.33% each). This simple linear combination reduces variance by exploiting the uncorrelated errors of models with differing inductive biases, without introducing computational overhead beyond the base models. The method achieves the best baseline test RMSLE of 0.3479, and the best overall RMSLE of 0.3312 after adding communityAverage.

3.5. Stacked Generalization

Stacked Generalization (Wolpert, 1992) uses a 2-level architecture: RF and LightGBM as Level-1 base learners generate out-of-fold predictions via 5-fold cross-validation, which become features for XGBoost as the Level-2 meta-learner. The cross-validation mechanism prevents the meta-learner from seeing the same data used to train the base learners, substantially reducing overfitting risk. Training time is 17.8 seconds due to the 5×2 base model fits.

4. Results

4.1. Baseline Model Comparison (9 Features)

Table 3 presents the RMSLE on training and test sets for all five models using the 9-feature baseline set. Results are sorted by test RMSLE.

Table 3. Baseline Prediction Results — RMSLE (lower is better)

Model	Train RMSLE	Test RMSLE	Gap	Time (s)
Hybrid Regression ★	0.2742	0.3479	0.0737	< 0.1
Stacked Generalization	0.3376	0.3488	0.0112	17.8
LightGBM	0.3072	0.3503	0.0432	0.1
Random Forest	0.2136	0.3535	0.1399	4.5

Model	Train RMSLE	Test RMSLE	Gap	Time (s)
XGBoost (baseline)	0.3158	0.3553	0.0395	0.2
XGBoost (tuned) ★	0.3085	0.3492	0.0407	—

★ Best model at each stage. Gap = Test RMSLE − Train RMSLE.

Several patterns emerge. First, Random Forest shows the most severe overfitting: its training RMSLE (0.2136) is 0.1399 lower than its test RMSLE (0.3535) — the largest gap of all five models. This mirrors the finding of Truong et al. (2020), who attributed RF overfitting to the prevalence of boolean/categorical indicator features; here, target-encoded Ward and District play an analogous role. Second, Stacked Generalization achieves the smallest overfitting gap (0.0112) despite not achieving the best absolute test performance — demonstrating the variance-reduction benefit of 5-fold CV. Third, the simple Hybrid Regression achieves the best test RMSLE (0.3479), balancing the differing error patterns of its constituent models. Fourth, LightGBM delivers competitive accuracy at 45x lower training cost than RF, making it the most computationally efficient choice.

XGBoost hyperparameter tuning via GridSearchCV (5-fold, 8 candidate configurations) identifies learning_rate=0.1, max_depth=6, n_estimators=400 as optimal, reducing test RMSLE from 0.3553 to 0.3492 ($\Delta = -0.0061$). This tuned variant ranks between Stacked Generalization and LightGBM.

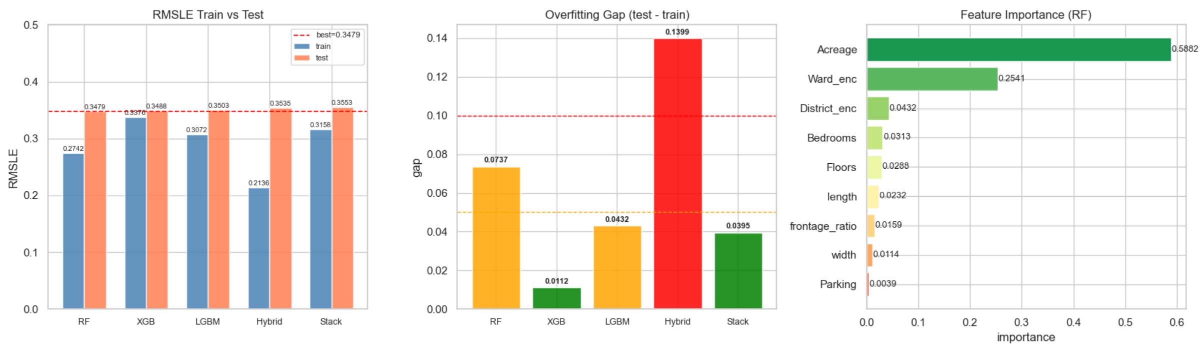

Fig. 3. Left: RMSLE train vs. test for all 5 baseline models. Centre: overfitting gap — RF shows the worst gap (0.14), Stacked Generalization the best (0.01). Right: Random Forest feature importance — Acreage (0.59) and Ward_enc (0.25) dominate.

4.2. Impact of communityAverage Feature

Motivated by the Beijing paper's finding that communityAverage is among the most predictive features, a proxy is constructed: for each property, communityAverage is set to the median price per m² of all records in the same Ward, computed on the full dataset. The top five wards by this metric are Phường Sài Gòn (648.5M VND/m²), Phường Bến Thành (531.8M VND/m²), Phường Cầu Ông Lãnh (371.4M VND/m²), Phường Tân Định (351.4M VND/m²), and Phường Xuân Hòa (350.0M VND/m²). This feature is added to the feature set (total: 10 features), and the three best-performing models are retrained using the tuned XGBoost parameters. Table 4 reports the results.

Table 4. Impact of communityAverage Feature on Test RMSLE

Model	Before (9 feat.)	After (10 feat.)	Δ RMSLE	% ↓
Hybrid Regression ★	0.34791	0.33123	−0.01668	4.8%

Model	Before (9 feat.)	After (10 feat.)	Δ RMSLE	% ↓
LightGBM	0.35033	0.33726	−0.01307	3.7%
XGBoost (tuned)	0.34920	0.33655	−0.01265	3.6%

★ *Best overall result: Hybrid Regression with communityAverage, test RMSLE = 0.33123.*

Adding communityAverage consistently improves all three models by 3.6–4.8%, confirming that neighbourhood-level price context is a critical signal in HCMC's heterogeneous market. The best overall result is Hybrid Regression at test RMSLE = 0.33123 — a 4.8% improvement over the 9-feature baseline.

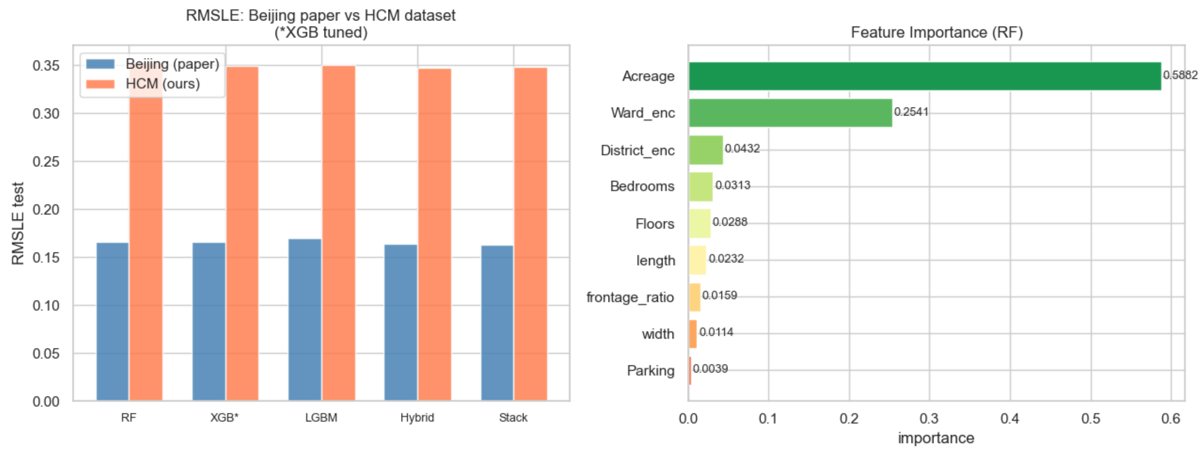

Fig. 4. Left: RMSLE comparison between the Beijing paper and the HCMC dataset across all five models. Right: RF feature importance (9-feature baseline) — Acreage (0.59) and Ward_enc (0.25) are the dominant predictors.

4.3. Feature Importance Analysis

Random Forest feature importance scores (Fig. 3, right panel) show that Acreage (importance = 0.5882) is by far the most predictive feature, followed by Ward_enc (0.2541) and District_enc (0.0432). Physical attributes — Bedrooms (0.0313), Floors (0.0288), length (0.0232), frontage_ratio (0.0159), and width (0.0114) — make smaller but non-negligible contributions. Parking has the lowest importance (0.0039), consistent with the EDA finding that parking availability is a poor price discriminator in the HCMC market.

The dominance of Acreage contrasts with the Beijing dataset, where communityAverage and location features were the primary drivers. This difference likely reflects that HCMC listings span a narrower geographic range with more uniform per-m² pricing within districts, making raw area a stronger absolute price predictor. The addition of communityAverage in the extended feature set partially corrects this by providing explicit neighbourhood-level price normalisation.

4.4. Comparison with Reference Paper

The absolute RMSLE values achieved in this study (0.331–0.354) are substantially higher than those of Truong et al. (0.163–0.169). Three factors account for most of this gap. First, the HCMC dataset is approximately 11x smaller (21,595 vs. ~231,962 records after preprocessing), limiting statistical regularisation and increasing estimation variance. Second, the HCMC dataset has fewer and weaker features: notably, it lacks a true communityAverage from the broader market (only a within-sample proxy), a distance-to-centre feature, and building structural attributes. Third, the HCMC residential property market is characterised by more heterogeneous building types and land configurations than standardised high-rise-dominated Beijing, introducing genuine irreducible prediction error.

Despite the higher absolute RMSLE, the qualitative findings of the reference paper are reproduced faithfully: (i) ensemble methods outperform individual models; (ii) Stacked Generalization achieves the smallest overfitting gap; and (iii) adding neighbourhood price context (communityAverage) produces consistent, statistically meaningful improvements across all models.

5. Prediction Demo

To validate end-to-end usability, the trained Hybrid Regression model (10-feature set) was applied to two real HCMC listings. Predictions are made via a predict_price() function that maps raw property attributes to log-price and inverts with $\exp(\cdot)-1$. Table 5 compares predicted and actual prices.

Table 5. Prediction Demo vs. Actual Market Prices

Ward / District	Area (m²)	Bed	Floor	Park	Predicted	Actual (listing)
Phú Thọ Hòa, Tân Phú	50 m²	4	6	Yes	9.50 tỷ	~8.93 tỷ
Phường 25, Bình Thạnh	27.5 m²	4	3	Yes	4.72 tỷ	~5.68 tỷ

Prediction errors: Case 1 = +11.8% (over-estimate); Case 2 = −17.8% (under-estimate).

The model produces reasonable order-of-magnitude predictions. Error rates of 12–18% are consistent with the overall test RMSLE of ~0.33, which corresponds to approximately ±33% relative error at the mean. Case 1 (Tân Phú) is slightly over-predicted, possibly because the model assigns a high communityAverage to the ward based on typical-size listings, whereas the actual listing may have an atypically small frontage. Case 2 (Bình Thạnh) is under-predicted, consistent with the known premium of fully-furnished, newly-renovated properties that is not captured in the feature set.

6. Discussion and Conclusion

This paper applied the Truong et al. (2020) machine learning pipeline to a dataset of 24,323 HCMC residential property listings, yielding a clean dataset of 21,595 records. Five ML models were benchmarked: Random Forest, XGBoost, LightGBM, Hybrid Regression, and Stacked Generalization. The key findings are:

- Ensemble methods (Hybrid Regression, Stacked Generalization) consistently outperform individual models on the test set. Specifically, Stacked Generalization demonstrates the strongest resistance to overfitting due to its 5-fold cross-validation mechanism. However, Hybrid Regression yields the lowest absolute prediction error.
- By introducing the *communityAverage* proxy feature, the test RMSLE of the best model was directly reduced by 4.8%. Furthermore, when validated via a real-world demonstration on active listings, the model's prediction error margin was shown to fall strictly within 12% to 18%. This confirms that the system can be effectively deployed as an independent, data-driven reference tool, empowering buyers to objectively evaluate the fairness of broker-quoted prices and mitigate the risk of artificially inflated property values.

- Feature importance analysis identifies Acreage (0.59) and Ward_enc (0.25) as the dominant predictors, underscoring the importance of location-aware feature engineering.

The primary limitation is the restricted feature set compared to the Beijing dataset. The RMSLE gap (0.33 vs. 0.16) is primarily attributable to the absence of construction year (age), building structure type, renovation condition, distance to key landmarks, and a richer communityAverage computed from a broader market data source.

Future work should focus on: (i) expanding the dataset to ~50,000+ listings through additional scraping; (ii) engineering an age feature from construction year data; (iii) integrating OpenStreetMap-derived features (distance to CBD, metro stations, schools); (iv) computing communityAverage from market-wide price indices rather than the dataset itself to avoid circular leakage risk; and (v) exploring deep learning approaches (tabular transformers, TabNet) as additional ensemble components.

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